

JESSICA NATHANIA

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PROFESSIONAL SUMMARY

I am a final-year Informatics Engineering student (GPA 3.99, Cum Laude) who thrives at the intersection of data, networks, and real-world problem solving. From building end-to-end data pipelines and dashboards using SQL, Python, Tableau, and Power BI, to administering Linux servers and managing network infrastructure. I bring both the technical depth and operational mindset that engineering environments demand. My attention to detail was sharpened through Cyber Security training in Malaysia, and my analytical instincts are proven through hands-on projects in FMCG pricing intelligence and retail sales analytics. Beyond data and networks, web development is both my craft and my passion. I have built production-grade applications and taken on freelance projects, turning ideas into fully functional websites. BNSP-certified in Network Administration, I don't just analyze problems. I solve them.

EDUCATION

Bachelor of Applied Technology in Informatics Engineering

Aug 2023 – Present

Politeknik Caltex Riau, Pekanbaru, Indonesia

- **GPA: 3.99 / 4.00** — Cum Laude | Top #1 by GPA, Informatics Engineering Program (Semesters I–VI consecutively)
- **Relevant Coursework:** Data Engineering, Business Analytics, Database Systems, Data Structures & Algorithms, Computer Networking, Network Security, Network Configuration & Administration (Cisco, MikroTik)
- **Research:** SawitMaju - Edge-AI Android Application utilizing YOLOv11 for real time oil palm ripeness detection and automated harvest data logging.

International Student Exchange — Cyber Security

October 2024

Politeknik Mersing Johor, Malaysia

- Coursework: Web Security, Digital Forensics, Network Security, Cryptography, Binary Exploitation
- Achieved 2nd Place — Mini Capture The Flag (CTF) Competition, applying log forensics and vulnerability analysis under time-constrained competition conditions

EXPERIENCE & TECHNICAL PROJECTS

Mie Sedaap vs Indomie: Market Dominance & Revenue Strategy | Python · Streamlit · Vercel

2026

- Built a pricing intelligence dashboard tracking FMCG market trends and promotions.
- Extracted, cleaned, transformed, and Visualized retail data using Python and SQL.

Global Malnutrition Data Warehouse | ETL Pipeline · Tableau Visualization · Embedded Website

2025

- Designed a Data Warehouse star-schema to support multi-dimensional analytics.
- Built end-to-end ETL pipelines to clean and load heterogeneous datasets.
- Delivered interactive Tableau dashboards visualizing malnutrition trends across 50+ countries.
- Optimized query speeds using advanced SQL indexing and schema normalization.

Superstore Sales & Customer Behavior Dashboard | ETL Pipeline · Tableau Visualization

2026

- Designed a Tableau dashboard analyzing revenue trends and customer segmentation, executing the full data pipeline from collection and cleaning to visualization.

E-Commerce Commercial Performance & Pricing Dashboard | Excel · KPI · Dynamic Slicers

2026

- Developed an Excel dashboard tracking e-commerce sales KPIs, conversions, and competitor pricing through large-scale transactional data cleaning and visualization.

Big Data & Distributed Computing Pipeline | Hadoop · Spark · Hive · Docker · MongoDB

2025

- Deployed a containerized multi-node cluster (Hadoop, Hive, Spark) using Docker Compose.
- Processed large-scale datasets with high accuracy using MapReduce and Hive queries.
- Ingested and queried semi-structured data using MongoDB and Apache Drill.

KaryaNusa — Full-Stack E-Commerce Application | Laravel · MySQL · Forecasting

2026

- Developed a website with business needs for Pekanbaru MSMEs.

- Designed a normalized MySQL database architecture, optimizing heavy JOIN queries to a 10ms response time.
- Validated and debugged a RESTful API through Postman using JSON payloads and HTTP code.

JobLinker Portal — Full-Stack Web Development | React · node js · MySQL

2024

- Developed a full-stack job portal using React to simplify recruitment.
- Developed a scalable RESTful API and optimized the database schema for job posting and application tracking.

Politeknik Caltex Riau Multipurpose Hall Reservation System | Laravel · MySQL

2024

- Developed a web-based reservation system (Laravel) to simplify the campus facility booking workflow.
- Implemented a scheduling feature with real-time availability to prevent double-bookings.
- Designed an efficient database to securely manage tenant applications and rental records.

SawitMaju – Oil Palm Ripeness Detection Mobile App | YOLOv11 · Android · Kotlin · Firebase

2025

- Developed an Edge-AI Android application using YOLOv11 for real-time oil palm fruit ripeness detection directly on-device, eliminating the need for internet connectivity in field conditions.
- Automated harvest data logging upon detection, enabling farmers to record and track harvest readiness efficiently through a mobile-first interface built in Kotlin.

Reservasi Kost Putra – Boarding House Reservation Platform | React · Node.js · MySQL

2025

- Built a full-stack boarding house reservation platform with real-time room availability, tenant application management, and scheduling to prevent double-bookings.
- Designed a normalized MySQL database schema and RESTful API backend (Node.js) consumed by a React frontend for seamless tenant and admin workflows.

SIG Vegetasi Pekanbaru – Urban Vegetation GIS Dashboard | React · Vite · GIS

2025

- Developed an interactive Geographic Information System (GIS) web dashboard visualizing urban vegetation distribution across Pekanbaru using React + Vite.
- Integrated geospatial data layers to enable spatial analysis of green coverage, supporting environmental monitoring and city planning decision-making.

Junior System Administrator | Linux servers · Docker · Networking · Cisco

2025

- Administered Linux servers and managed Docker deployments for academic workloads.
- Monitored system logs to detect anomalies and documented troubleshooting workflows.
- Authored and maintained version-controlled internal SOPs for server configuration.
- Managed infrastructure changes and rollback runbooks using Git.

Campus Network Topology Design | BNSP Certification Case Study: Politeknik Caltex Riau

2024

- Designed a scalable and comprehensive campus network topology case study for Politeknik Caltex Riau to fulfill BNSP certification requirements.
- Planned and simulated IP addressing, VLANs, routing, and switching architectures.
- Architected a network structure focused on secure, reliable, and efficient inter-departmental communication.

TECHNICAL SKILLS

- **Database & Query:** SQL (Advanced), MySQL, MongoDB, Relational Schema Design, Query Optimization
- **Data Visualization:** Tableau, Excel (Advanced), Google Sheets, Dashboard Design
- **Data Engineering:** ETL Pipeline Design, Data Warehousing, Star Schema Modeling, Batch Processing, Data Quality
- **Big Data & Distributed:** Hadoop (HDFS, MapReduce), Apache Spark, Hive, HBase, Apache Drill
- **IT Documentation:** SOP Authoring, System Log Analysis, Troubleshooting Documentation, Markdown
- **Programmer :** Laravel, React, RESTful API, Postman, JWT Auth, [Node.js](#), Java
- **Networking & Infrastructure:** Junior Network Admin (BNSP Certified), Linux Server, Docker, Git, VPS, TCP/IP
- **Security:** Cyber Security Fundamentals, CTF (Log Analysis, Digital Forensics), JWT Authentication
- **Languages:** Bahasa Indonesia (Native), English (Professional Working Proficiency)

CERTIFICATIONS

- *Business Analytics with Excel - Microsoft - SkillUP* Juni 2026
- *Data Analysis with Python — IBM [ID:DA0101EN]* Juni 2026

- *Big Data 101 - IBM - Cognitive Class* *Juni 2026*
- *Data Science Methodology - Cognitive Class* *Juni 2026*
- *Business Intelligence Fundamentals - IBM -SkillUP* *Juni 2026*
- SQL (Advanced) Certification — HackerRank, [ID: B797F73OFECB] *April 2026*
- Software Engineer Role Certification — HackerRank, [ID: 3E1E3AD1C397] *April 2026*
- REST API (Intermediate) Certification — HackerRank, [ID: 85AE17222D23] *April 2026*
- Go (Basic) Certification — HackerRank [ID: 524575FA19E3] *Apr 2026*
- Developing Back-End Apps with Node.js and Express — IBM / Coursera [ID: GO0H26GYGFMV] *June 2025*
- Generative AI Basics (AWS) — Dicoding Indonesia *June 2025*
- Cloud Security Essentials (CSE) — EC-Council *September 2025*
- *Red Hat System Administration I (RH124) — Red Hat* *Aug 2024*
- MikroTik Certified Network Associate (MTCNA) — MikroTik *July 2025*
- Junior Network Administrator Certification — BNSP (Badan Nasional Sertifikasi Profesi) *2025*

ACHIEVEMENTS

- **Top 1 Best IT Student** — Informatics Engineering Program, Politeknik Caltex Riau *2023 – Present*
- **1st Place** — Best Website Project Competition, Pekanbaru *(2024)*
- **2nd Place** — Mini Capture The Flag (CTF) Competition, Politeknik Mersing Johor, Malaysia *(2024)*
- Regional Government Academic Excellence Scholarship — Riau Province *(2024 – Present)*
- Research Journal Contributor — IT Risk Management & Technology Trends *(2025)*
- Research Journal Contributor — YOLOv8 Implementation for Oil Palm Fruit Ripeness Detection *(2025)*

LEADERSHIP, EVENTS & ORGANIZATIONS

- President / Chairman — TECH EXPO III, Pekanbaru: Increased event participation by 50% through interactive robot, IoT, Web, and BI showcases
- President / Chairman Research & Technology Dept. — ITSA: Organized workshops, seminars, and technology events for students
- Head of Training & Development Division — CSIRT: Responsible for designing and delivering cyber security training programs and skill development workshops for members
- President / Chairman — Cyber Security Workshop: Led the organization and execution of a campus-wide cyber security workshop
- Master of Ceremonies — Basic Leadership Training, Student Executive Board (SEB): Managed event flow, introduced speakers, and maintained audience engagement throughout the training program
- Master of Ceremonies — Enabling Skill: Public Speaking, Politeknik Caltex Riau: Hosted the event and engaged 500+ attendees
- Public Relations — ITSA Farewell Party Generasi'19: Developed partnerships and sought out sponsorships for the event
- Public Relations — Seminar by CSIRT 2024: Coordinated external relations and served as information coordinator
- Treasurer — Backend CTS Workshop: Managed financial planning, budgeting, and expense tracking for the workshop